Syllabus

Course code	CSE 305	**Course name**	Computer Architecture	**credits**	3
Instructor	Name : Hyokeun Lee		Homepage : https://relacslab.github.io/		
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	Office hour : appointment is required				
1. Goals	Like skyscrapers, well-architecting a system that incorporates delicate, bleeding-edge technologies is pressing. This lecture covers the basics of computer architecture, including pipeline designs, memory managements, and some issues related to caches (e.g., policies and coherence). There will be ***two*** term projects based on an RTL design and an architecture simulator.				
2. Textbooks	**Main**: Patterson and Hennessy, "Computer Organization and Design (MIPS Edition)" **Sub**: Research Papers				
3. Prerequisites	▪ Logic Design (e.g., gates, flip-flops...) ▪ Digital System Design (e.g., Verilog HDL) ▪ C/C++ programming-related classes				

4. Ratings (%)	Attendance	Homework	Mid-term	Final-term	Project	Others	Overall
	5	0	30	35	30	0	100

5. Agenda	**Week**	**Contents**
	1	Introduction to Computer Architecture
	2	Instruction Set Architecture (1)
	3	Instruction Set Architecture (2)
	4	Single- & Multi-Cycle Microarchitectures
	5	Pipelined Microarchitecture: Introduction
	6	Pipelined Microarchitecture: Data Hazard and its Handling
	7	Pipelined Microarchitecture: Control Hazard and its Handling
	8	Mid-Term Exam
	9	Advanced Microarchitectures
	10	Memory Systems: Cache (1)
	11	Memory Systems: Cache (2)
	12	Memory Systems: Cache (3)
	13	Memory Systems: Evolution of Virtual Memory
	14	IO Device and Its Managements
	15	Final-Term Exam

6. Notes for students	▪ F will be given if cheating is caught no matter what the case is ▪ One grade lower if not taking either mid-term or final-term exam ▪ This class has a Verilog project